DATA SCIENCE & ARTIFICIAL INTELLIGENCE & CS/IT



Calculus and Optimization

Mean Value Theorems

One Shot



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

DERIVATIVES & THEIR TYPES

(PART-3)

Remaining Portion :— JACOBIAN

Topics to be Covered



Topic MEAN VALUE THEOREMS

1. Lagrange's M.V.Th. of Differentials
2. Rolle's M.V.Th.
3. Cauchy's M.V.Th.
4. L-M.V.Th for Integrals.

HW If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{zx}\right)$ then $x^2u_x + y^2u_y + z^2u_z = ?$

γ β

Ans = 0

Sol: $u = f(\gamma, \beta)$ where $\gamma = \frac{1}{x} - \frac{1}{y}$ & $\beta = \frac{1}{x} - \frac{1}{z}$

ie $u \longrightarrow (\gamma, \beta) \longrightarrow (x, y, z)$

$$u_x = \frac{\partial u}{\partial x} = \frac{\partial u}{\partial \gamma}\left(\frac{\partial \gamma}{\partial x}\right) + \frac{\partial u}{\partial \beta}\left(\frac{\partial \beta}{\partial x}\right) = u_\gamma\left(-\frac{1}{x^2}\right) + u_\beta\left(-\frac{1}{x^2}\right)$$

$$u_y = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial \gamma}\left(\frac{\partial \gamma}{\partial y}\right) + \frac{\partial u}{\partial \beta}\left(\frac{\partial \beta}{\partial y}\right) = u_\gamma\left(+\frac{1}{y^2}\right) + u_\beta(0)$$

$$u_z = \frac{\partial u}{\partial z} = \frac{\partial u}{\partial \gamma}\left(\frac{\partial \gamma}{\partial z}\right) + \frac{\partial u}{\partial \beta}\left(\frac{\partial \beta}{\partial z}\right) = u_\gamma(0) + u_\beta\left(+\frac{1}{z^2}\right)$$

$$\Rightarrow \begin{cases} x^2u_x = -u_\gamma - u_\beta \\ y^2u_y = u_\gamma \\ z^2u_z = u_\beta \end{cases}$$

$$x^2 u_x + y^2 u_y + z^2 u_z = u_\delta + u_\beta - u_\alpha - u_\gamma$$

$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$$

Type II **JACOBIAN**: If $u = u(x,y,z)$, $v = v(x,y,z)$, $w = w(x,y,z)$

i.e. $(u,v,w) \longrightarrow (x,y,z)$

then **Derivative** of (u,v,w) with respect to (x,y,z) is called **Jacobian** & it is defined as

$$J = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

Note (1) If $(u,v) \longrightarrow (x,y)$ then $J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$

Also evaluate $\frac{\partial(x,y)}{\partial(u,v)} = ? = \frac{1}{\partial(u,v)/\partial(x,y)} \Rightarrow J' = \frac{1}{J}$ i.e. $\boxed{JJ' = 1}$

where $J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ & $\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = J'$

i.e. $\boxed{JJ' = 1}$

(2) if $(u,v) \longrightarrow (r,s) \longrightarrow (x,y)$ then $\frac{\partial(u,v)}{\partial(x,y)} = ?$

$$J = \boxed{\frac{\partial(u,v)}{\partial(x,y)} = \frac{\partial(u,v)}{\partial(r,s)} \cdot \frac{\partial(r,s)}{\partial(x,y)}}$$

it is called Chain Rule of Jacobian.

(3) **Dependent func^n** :- Two func^n f & g are called dependent if ∃ a mathematical Relationship b/w them. which should be free from Ind Variables

Independent func^n :- f & g are called Ind if there DNE any mathematical Relationship b/w them.

Shortcut: let $f=f(x,y)$ & $g=g(x,y)$ i.e $(f,g)\longrightarrow(x,y)$

s.t $J=\dfrac{\partial(f,g)}{\partial(x,y)}=0 \Longrightarrow f$ & g are defendent func^n

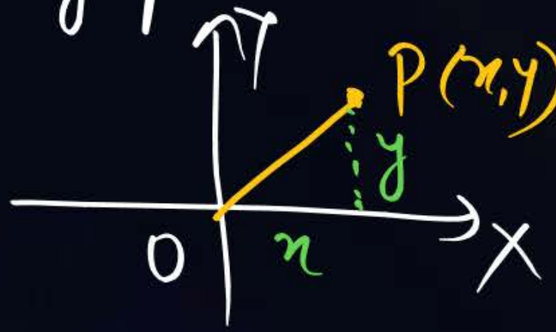
& if $J=\dfrac{\partial(f,g)}{\partial(x,y)}\neq 0 \Longrightarrow$ ∴ ∴ Independent func^n

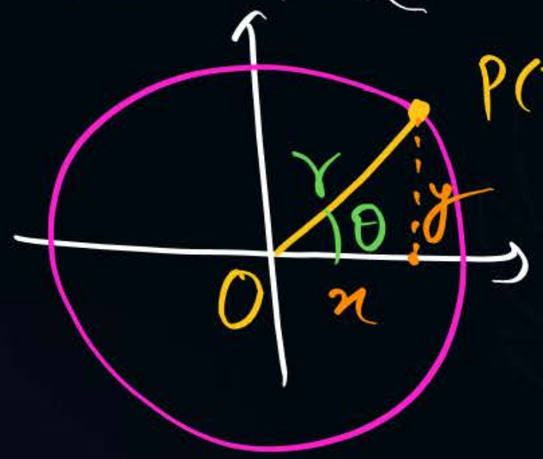
Q: If $x = r\cos\theta$, $y = r\sin\theta$ then evaluate $\frac{\partial(x,y)}{\partial(r,\theta)} = ?$

OR

Find the Jacobian of polar Coordinates, (r, θ)

Note: If $P(x,y)$ & $P(r,\theta)$ are the Cartesian and Polar Coordinates of any point that lies on Circle then





$\frac{x}{r} = \cos\theta$

$\frac{y}{r} = \sin\theta$

$\Rightarrow$ $x = r\cos\theta$, $y = r\sin\theta$

where $r^2 = x^2 + y^2$, $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$x = r\cos\theta,\ y = r\sin\theta$

$\because (x,y) \longrightarrow (r,\theta)$ then

$$J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta$$

$= r$ Ans

Note: If $x = r\cos\theta,\ y = r\sin\theta$ then $\frac{\partial(x,y)}{\partial(r,\theta)} = r$ i.e $\neq 0$

Here $(x,y) \longrightarrow (r,\theta)$

So x & y are Independent func^{n}.

But we have Relationship b/n x & y as follows; $x^2 + y^2 = r^2$

Take care, it is a Relationship Not free from Ind Variable



Q: if $u = x^2 - y^2$, $v = 2xy$ then $\frac{\partial(u,v)}{\partial(r,\theta)} = ?$ where Notations have their usual meaning.

Sol: w.k. that $x = r\cos\theta$ & $y = r\sin\theta$ i.e

$$(u,v) \longrightarrow (x,y) \longrightarrow (r,\theta)$$

So $$\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \cdot (r)$$

$$= \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} \cdot (r) = 4(x^2+y^2) \cdot r$$

$$= 4(r^2) \cdot r = 4r^3 \quad \text{Ans}$$

Q: if $u=\tan^{-1}x-\tan^{-1}y$ & $v=\frac{x-y}{1+xy}$ then Prove that these two functions are Dependent. if ∃ a Relation ship b/n them.

Sol: (M-I) using observations :-

$u=\tan^{-1}x-\tan^{-1}y$

$u=\tan^{-1}\left(\frac{x-y}{1+xy}\right)$

$u=\tan^{-1}(v)$

$\tan u=v$ or $\boxed{v=\tan u}$

(M-II) $(u,v)\to(x,y)$ so

$$J=\frac{\partial(u,v)}{\partial(x,y)}=\begin{vmatrix}\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y}\\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y}\end{vmatrix}$$

$$=\begin{vmatrix}\frac{1}{1+x^2} & \frac{-1}{1+y^2}\\ \frac{1+y^2}{(1+xy)^2} & \frac{-1-x^2}{(1+xy)^2}\end{vmatrix}\Rightarrow =\cdots=0$$

if u & v are Dependent

Mean Value Theorems (New chapter)

Note:

① Slope of line passing through (x_1, y_1) & (x_2, y_2) is $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

② In general, slope of tangent at any Random point is

$$m = \tan\theta = f'(x)$$

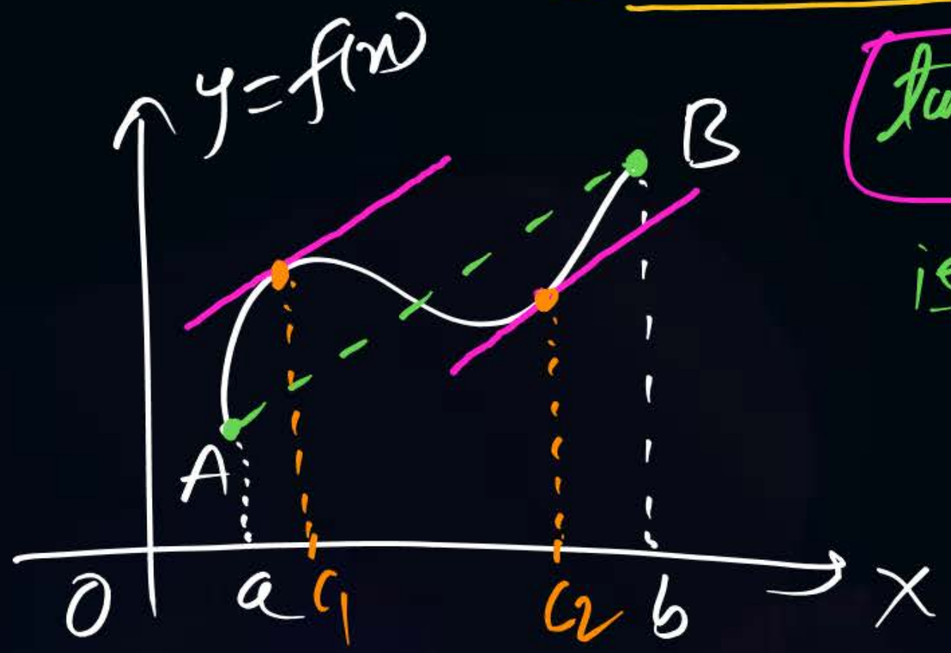
③ Slope of Horizontal line is $m = \boxed{f'(x) = \tan 0 = 0}$

④ **Elementary func^n** – All polynomial func^n, Exp func^n, log func^n, Trig. Functions & Inverse Trig func^n are called E-Functions.

⑤ All E-func^n are Continuous as well as Differentiable in their Domain.

(1) Lagrange's M.V Th

Let $f(x)$ is defined in $[a,b]$ s.t (i) $f(x)$ is Cont in $[a,b]$, (ii) $f(x)$ is Diff in (a,b) then ∃ at least one c in (b/w) a & b for which



$A(a, f(a))$, $B(b, f(b))$

Tangent at c is $\parallel^{gl}$ to chord AB

i.e Slope of tangent = Slope of AB

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

Note:- Converse of LMVTh is not necessarily True.

(2) Rolle's M.V.Th →

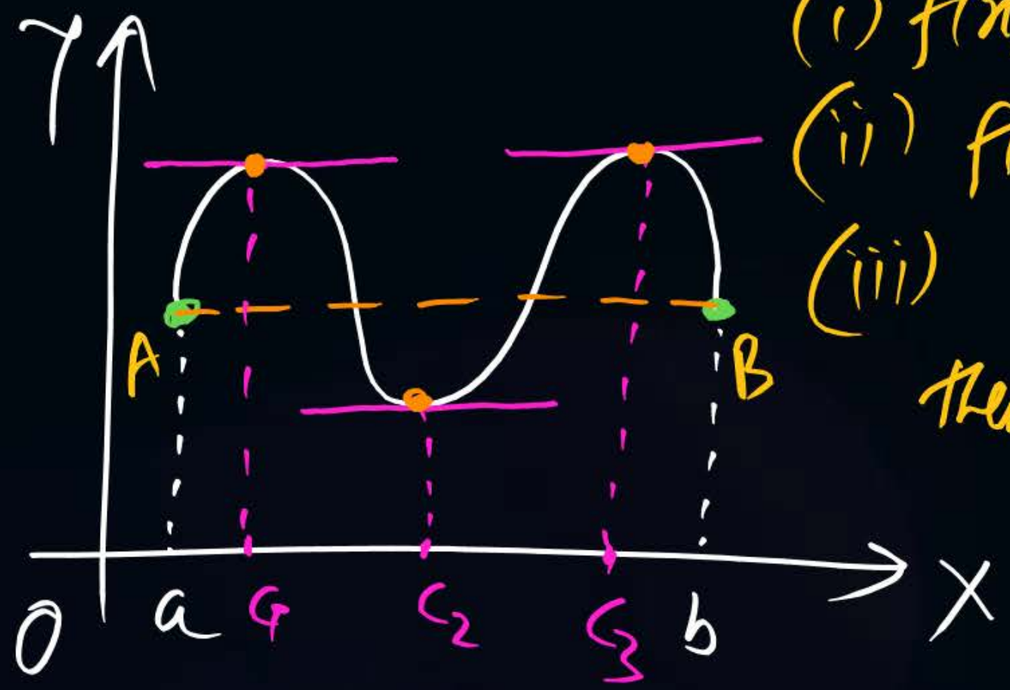
Let $f(x)$ is defined in $[a,b]$ s.t

(i) $f(x)$ is continuous in $[a,b]$

(ii) $f(x)$ is Diff in (a,b)

(iii) $f(a)=f(b)$

then $\exists$ at least one point c in (b/w) a & b for which



tangent is Horizontal

or tangent is $\perp^r$ to X axis

or $\boxed{f'(c)=0}$

(3) **Cauchy's M.V.Th** :- Let $f(x)$ & $g(x)$ are two func^n defined in $[a,b]$ s.t

(i) Both $f(x)$ & $g(x)$ are continuous in $[a,b]$

(ii) " " " " differentiable in (a,b)

(iii) $g'(x) \neq 0 \; \forall \; x \in (a,b)$

then $\exists$ at least one c in (b/n) a & b for which

$$\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$$

(Ratio of Difference of $f(x)$ & $g(x)$ at a & b) = (Ratio of slope of tangent of $f(x)$ & $g(x)$ at c)

Q: If $f(x)=x^2$ & $g(x)=x^3$ defined in $[1,2]$ then verify C.M.V.Th. and hence evaluate $c=?$

Sol:

(i) Both $f(x)$ & $g(x)$ are Continuous in $[1,2]$

(ii) " " " " Differentiable in $(1,2)$

} ∵ Both are E-func^n.

(iii) $g'(x)=3x^2$, Now $g'(x)\neq 0 \ \forall \ x\in(1,2)$

i.e. $f(x)$ & $g(x)$ are satisfying all the conditions of C.M.V.Th Hence,

$f(x)=x^2 \Rightarrow f(1)=1,\ f(2)=4$

$f'(x)=2x \Rightarrow f'(c)=2c$

$g(x)=x^3 \Rightarrow g(1)=1,\ g(2)=8$

$g'(x)=3x^2 \Rightarrow g'(c)=3c^2$

By C.M.V.Th,

$$\frac{f(2)-f(1)}{g(2)-g(1)} = \frac{f'(c)}{g'(c)}$$

$$\frac{4-1}{8-1} = \frac{2c}{3c^2}$$

$$\frac{3}{7} = \frac{2}{3c}$$

$$c = \frac{14}{9}$$

$\therefore c = \frac{14}{9} \in (1,2)$

Hence C.M.V.Th is verified.

Q: If $f(x)=\log_e x$ & $g(x)=\log_e\left(\frac{1}{x}\right)$ defined in $[1,2]$ then By C.M.V.Th, $c=$?

(a) 1.0

(b) 1.25

(c) 1.5

(d) Any Value b/n 1 & 2

Sol: $f(x)=\log x$, $g(x)=\log(x^{-1})=-\log x$

Both $f(x)$ & $g(x)$ are cont as well as diff in $(1,2)$

(∵ these are also E-func^n)

$\Rightarrow f'(x)=\frac{1}{x}$, $g'(x)=\frac{-1}{x}$, $a=1$, $b=2$

∵ $g'(x)\neq 0\ \forall\ x\in(1,2)$ So By C.M.V.Th,

$$\frac{f(2)-f(1)}{g(2)-g(1)}=\frac{f'(c)}{g'(c)} \Rightarrow \frac{\ln 2-0}{-\ln 2-0}=\frac{1/c}{-1/c} \Rightarrow -1=-1$$

is identity

i.e theorem is true for any value of c So Ans is (d)

Q. Verify R.Th for $f(x)=\sin^4 x+\cos^4 x$ in $\left[0,\frac{\pi}{2}\right]$ & Hence find c = ?

(a) 0 — f(x) is Cont as well as Diff (∵ it is a combination of E-func^n)

(b) $\pi/2$ — Now $f(0)=0+1^4=1$ & $f(\pi/2)=1^4+0=1$

(c) ✓ $\pi/4$ — i.e. $f(0)=f(\pi/2) \Rightarrow$ f(x) satisfies all the conditions of R.Th.

(d) $\pi/6$ — Soo $\exists\ c \in (0,\pi/2)$ s.t $\boxed{f'(c)=0}$

$f(x)=\sin^4 x+\cos^4 x$

$f'(x)=4\sin^3 x\cdot\cos x+4\cos^3 x(-\sin x)$

$=-4\sin x\cos x(\cos^2 x-\sin^2 x)=-2\sin 2x\cdot\cos 2x=-\sin 4x$

$-\sin 4c=0$

$f'(c) = 0$

$-\sin 4c = 0$

$\sin 4c = \sin n\pi, \; n \in I$

$4c = n\pi$

$c = \frac{n\pi}{4}, \; n \in I$

$c = \ldots\ldots -\frac{\pi}{2}, -\frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \ldots\ldots$

i.e. $c = \pi/4$

Note:

(1) $\sin x = 0$

$\sin x = \sin n\pi \Rightarrow x = n\pi$

i.e. $x = \ldots -2\pi, -\pi, 0, \pi, 2\pi, 3\pi \ldots$

(2) $\cos x = 0$

$\cos x = \cos(2n+1)\frac{\pi}{2}$

$\Rightarrow x = (2n+1)\frac{\pi}{2}, \; n \in I$

$x = \ldots\ldots -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2} \ldots\ldots$

Q: Find the equⁿ of tangent at the point on the curve $f(x)=\sin^4 x+\cos^4 x$ in $[0,\pi/2]$ whose tangent is $\parallel^{r}$ to x axis ? & find coordinates of that point.

Sol: Let the point is $C(c, f(c))$ where tangent is $\parallel^{r}$ to x axis.

Now slope of tangent at C is $m = f'(c) = 0$

$$\Rightarrow -\sin 4c = 0 \Rightarrow c = \frac{\pi}{4}$$

Now $f(c) = f(\pi/4) = (\sin^4 x+\cos^4 x)_{x=\frac{\pi}{4}} = \left(\frac{1}{\sqrt{2}}\right)^4+\left(\frac{1}{\sqrt{2}}\right)^4 = \frac{1}{2}$

Hence equⁿ of tangent at $C\left(\frac{\pi}{4}, \frac{1}{2}\right)$ having slope $m=0$ will be,

$$y-\frac{1}{2} = m\left(x-\frac{\pi}{4}\right) \Rightarrow y=\frac{1}{2}$$ Ans & Coordinates $=\left(\frac{\pi}{4}, \frac{1}{2}\right)$

Q. Verify L.M.V.Th for $f(x)=x^{1/3}$ in $[-1,1]$ & hence evaluate $c=?$

Sol: $f'(x)=\frac{1}{3}x^{\frac{1}{3}-1}=\frac{1}{3x^{2/3}}$

But at $x=0$, $f'(0)=DNE$

i.e. $f(x)$ is Not Differentiable at every point in $[-1,1]$. Hence 2nd Condition of LMVT is Not Satisfied

i.e. LMVT is not applicable

(ii) 2nd part is senseless.

ANALYSIS :→ (PODCAST)

$f(x) = x^{1/3} \Rightarrow f'(x) = \frac{1}{3x^{2/3}}$ in $[-1, 1]$

Let's try to apply LMVT forcefully bw -1 & 1

ie $\frac{f(1)-f(-1)}{1-(-1)} = f'(c)$

$\frac{(1)^{1/3}-(-1)^{1/3}}{2} = \frac{1}{3c^{2/3}}$

$\frac{1-(-1)}{2} = \frac{1}{3c^{2/3}}$

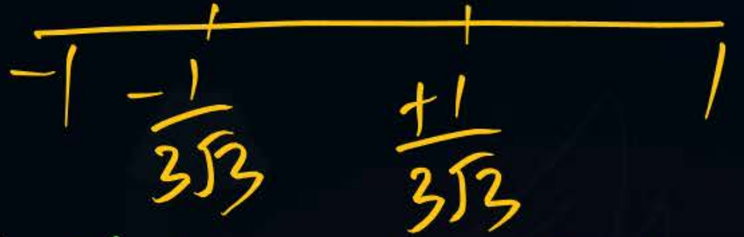
ie $3c^{2/3} = 1$

$c^{2/3} = \frac{1}{3}$

$(c^{2/3})^3 = \left(\frac{1}{3}\right)^3$

$c^2 = \frac{1}{27}$

$c = \pm\frac{1}{3\sqrt{3}}$



it is possible to find c But it does not imply that $f(x)$ is Cont & Diff both.

Remark:

LMVTh.

$$\text{If } [f(x) \text{ is cont \& Diff}] \Rightarrow \left[\exists\, c \in (a,b) \text{ s.t } \frac{f(b)-f(a)}{b-a} = f'(c)\right]$$

$$\Leftarrow \not$$

Qe Consider $f(x)=(x-1)^2+2$ then Verify LMVT in $[0,1]$ (Ans $c=\frac{1}{2}$)

OR

Qe Consider $f(x)=(x-1)^2+2$ then find Coordinates of that point on the curve $f(x)$ where tangent is $\parallel^{el}$ to the chord joining $(0,3)$ & $(1,2)$

OR

Ans $\left(\frac{1}{2}, \frac{9}{4}\right)$

Qe if $f(x)=(x-1)^2+2$ then find equ^{n} of tangent which is $\parallel^{el}$ to the chord joining $(0,3)$ & $(1,2)$.

Ans: $4x+4y=11$

OR

Qe for the func^{n} $y=(x-1)^2+2$, At least at one point in the interval $[0,1]$, the value of $\frac{dy}{dx}=?$ (Ans $=-1$)

THANK - YOU

Tel: drbunnet Sir pw